A Coverage-Controlled Diagnostic for In-Context Compositional Generalization of a Withheld Causal Mechanism

Nathan Peterson
Prysmos (independent)
orbitnate@gmail.com

June 2026

Abstract

We present a small, *coverage-controlled* diagnostic that tests whether a system can do one specific thing in-context: *infer the effect of a causal variable it was never shown a clean intervention for*, and compose that inferred effect with directly-observed ones to predict a novel joint intervention. The diagnostic has three design features that, to our knowledge, are not combined in existing in-context compositional-generalization or causal-foundation-model benchmarks. First, it **injects the known group operation** (modular addition over a finite cyclic group), so a failure cannot be blamed on failing to learn the combination rule—the only thing left to learn is the per-variable deconfounded read and its transport to a withheld variable. Second, it is **coverage-controlled and leak-free** by construction: the entire interventional block of one variable is withheld, and every prompt row that exactly matches a query is removed. Third, before any model is queried it runs a **$0 fairness gate**—the same analytic composer that defines the score ceiling is executed on *exactly* the prompt rows, and the run aborts if the answer is not recoverable. We apply the identical instrument across model scales. A frontier LLM (DeepSeek V4 Flash) reaches the analytic ceiling (+0.937, 12/12 queries exact) on 2- and 3-variable regimes; a from-scratch ≈74k-parameter transformer, given the *same* injected group structure, collapses to ≈ 0 even on the easy all-revealed case, and increasing training-task diversity does not fix it. We argue the contribution is the *instrument and its controls*, not the (already-documented) memorize-vs-infer phenomenon, and we are explicit about the limits: single seed, single LLM, an easy identifiable regime, and a diversity-threshold argument that is a-priori rather than fully empirical.

1 Introduction

A growing literature shows that transformers can compose skills in-context but often *memorize* rather than *infer* when the training distribution is narrow [1, 2], with a task-diversity threshold governing the transition [4, 5]. In parallel, "causal foundation models" learn to do amortized causal inference in-context [6, 7, 8, 9], but their generalization is typically evaluated over held-out *graphs*, *datasets*, or *distribution shift*—not over a held-out *mechanism*. We isolate a narrower, cleaner primitive:

> *Given confounded observational data and clean interventional ("forced") data for some variables but* ***none*** *for one withheld variable i^*, can a system infer i^*'s effect by cross-referencing the other interventions, and compose all effects to predict a novel joint intervention—with no side-channel revealing the answer?*

On terminology. We call this *compositional generalization of a withheld causal mechanism.* It is *not* Pearl–Bareinboim transportability (recovering an effect across domains related by a selection diagram); we do not invoke the transport formula, and we use the word "transport" only informally for "carry an inferred per-variable effect across interventional blocks."

Contribution. The contribution is a *measurement instrument*, not a new phenomenon. Its three load-bearing properties are: (i) **injecting the known combine operator** so a negative result cannot be attributed to failure to learn the operation; (ii) **coverage control and leak removal** via a whole-variable holdout plus exact-assignment exclusion; and (iii) a **$0 pre-flight fairness gate** that runs the analytic ceiling composer on the literal prompt rows and aborts unfair instances before any spend—so every reported number is fair on its exact prompt rows by construction. We then run the *same* instrument on a tiny from-scratch model and on a frontier LLM, yielding a clean scale contrast. We report honestly that the LLM passing an identifiable task is expected; the value is the instrument and the controls, and the contrast they make legible.

2 Related work

In-context skill composition and memorization. He et al. [1] show emergence of in-context skill composition in modular arithmetic; Kobayashi, Schug et al. [2] characterize when transformers compositionally generalize in-context. Both report a memorize-vs-compose distinction closely related to what our negative arm measures; our addition is the causal/coverage framing and the injected operator, not the phenomenon. Lippl & Stachenfeld [3] give a kernel theory for when compositional structure yields compositional generalization.

Task diversity thresholds. Raventós et al. [4] establish that a *pretraining task-diversity threshold* (on the order of 2^{14} tasks in their regression setting) governs the emergence of non-Bayesian in-context learning; Nguyen & Reddy [5] analyze the learning kinetics of the memorization-to-generalization transition. We discuss in §6 why injecting the operator changes the relevance of this threshold to our setting.

Amortized / in-context causal models. CSIvA [6] learns to induce causal structure; ACTIVA [7], Cond-FiP [8], and Do-PFN [9] amortize causal-effect estimation in-context. Montagna et al. [10] show amortized causal-discovery transformers remain bound by identifiability and do not transport off-distribution. GIM [11] generalizes to unseen intervention targets but with an informative side-channel; removing that side-channel is precisely our test. Abedsoltan et al. [12] study autoregressive compositional task generalization ($D \to D^T$). Across these, we did not find the specific slice—*infer a globally-withheld intervention effect, no side-channel, with the combine operator injected and a $0 pre-flight fairness gate*—and that gap is the instrument's claim to novelty (an absence-of-evidence claim; see §6).

3 The diagnostic

World. A world has n dials, each taking values in $\{0, \ldots, C-1\}$, and a meter $R \in \{0, \ldots, C-1\}$. Each dial i has a secret bijection (conversion table) g_i, drawn as a uniform random permutation of $\{0, \ldots, C-1\}$. The meter is additive over the cyclic group:

$$R \;=\; \left(\textstyle\sum_{i=1}^{n} g_i(d_i) \;+\; \eta_R\right) \bmod C, \tag{1}$$

where η_R is meter noise. A hidden state $H \sim \mathrm{Uniform}\{0, \ldots, C-1\}$ confounds the system: in observational data every free dial is $d_i = (H + \eta_i) \bmod C$ (so the dials are correlated through H and naive dial-vs-meter statistics mislead). Noise is sparse: each free dial value is replaced by a fresh uniform draw with probability 0.15 (else driven by H), and the meter is corrupted with probability 0.06. An intervention $\mathrm{do}(d_i = v)$ fixes dial i to v while the others remain free.

Prompt. The system is shown, in-context: (a) confounded *observational* rows; (b) clean *forced* (interventional) blocks $\mathrm{do}(d_i = v)$ for the *present* dials; and (c) *nothing* for one **withheld** dial i^*, whose entire forced block is removed. $i^* = \mathrm{hash}(\mathrm{seed}) \bmod n$ is fixed per world, chosen blind to outcomes. The query asks for R under a novel joint forcing $(\, d_{i^*} = v,\ d_j = w\,)$ with $v \neq \mathrm{ref}$ ($\mathrm{ref} = 0$), a combination never shown. Because i^*'s block is absent, its effect must be inferred *across* blocks—from how the meter shifts as i^* wanders inside another dial's forced block—then composed with the directly-read effects.

Analytic composer = information ceiling. A reference reasoner reconstructs the answer from the per-row (d, R) data only, never the answer key. For each present dial it recovers $g_i(v) - g_i(\mathrm{ref})$ as a relative mode-shift inside its forced block, conditioning the other dials to a common value w_0 so the shared background cancels. For the withheld dial it recovers $g_{i^*}(v) - g_{i^*}(\mathrm{ref})$ *cross-block*, from rows inside another dial's forced block where the still-free i^* takes v vs. ref. It anchors the constant $K = \sum_i g_i(\mathrm{ref})$ from all-reference rows, and predicts a point mass at $(K + \sum_i [g_i(a_i) - g_i(\mathrm{ref})]) \bmod C$. A conditioning cell is trusted only with ≥ 10 rows. This composer defines the achievable *ceiling*; because the meter is mildly noisy, a point mass cannot reach the diffuse oracle exactly, capping the ceiling in $[0.933, 0.938]$ (below 1).

Score. For a predicted distribution p and the oracle joint-do distribution q over $\{0, \ldots, C-1\}$, with u the uniform distribution, define the total-variation similarities $\mathrm{tv}(p) = 1 - \frac{1}{2}\|p - q\|_1$ and $U = \mathrm{tv}(u)$, and

$$\mathrm{score}(p) \;=\; \mathrm{clip}\Big(\frac{\mathrm{tv}(p) - U}{1 - U},\, -1,\, 1\Big). \tag{2}$$

An uninformed (uniform) prediction scores 0; an exact match scores 1 up to oracle diffuseness.

Leak controls. Two exclusions, both required: (1) the *whole-dial holdout* drops every forced block of i^* (so the effect cannot be obtained by elimination); (2) the *exact-assignment exclusion* drops every prompt row (observational or forced) whose dial values exactly match *any* query, closing a single-row lookup leak that a 2-dial design otherwise admits.

$0 pre-flight fairness gate. Before any model is queried, the run asserts that (a) i^* has zero forced blocks, (b) no remaining prompt row matches any query assignment, and (c) the analytic composer, executed on *exactly* the rows that go into the prompt, scores mean $\geq 0.80 \times$ ceiling and is 100% populated. If any check fails the instance is unfair and the run **aborts with no model call**. Because the composer consumes the same row object that renders the prompt, "the test was fair on the exact prompt rows" holds by construction for every reported number. This gate operationalizes a lesson from our own prior work, where an apparent effect turned out to be a coverage artifact (a model graded on cells it never sampled).

Regime	Ceiling (3-seed range)	Min. fair rows/block	Floor (strongest cheater)
2 dials, C=4	0.933–0.937	75	≈ 0
2 dials, C=5	0.934–0.937	150	≈ 0
2 dials, C=6	0.934–0.937	150	≈ 0
2 dials, C=7	0.934–0.936	150	≈ 0
2 dials, C=8	0.934–0.936	150	≈ 0
3 dials, C=3	0.936–0.937	300	≈ 0
3 dials, C=4	0.936–0.938	600	≈ 0
All regimes	0.933–0.938	—	—

Table 1: Feasibility/identifiability sweep (analytic composer; seeds 20260606–08). The ceiling is essentially constant across C and dial count; only the coverage budget grows. "Min. fair rows/block" is the smallest budget at which both the all-revealed and withheld ceilings are ≥ 0.70 and 100% populated on all three seeds.

Control	Result
1. Real analytic composer (positive control)	scores 0.934–0.939 across all 9 cells
2. Non-compositional cheaters fail (one-cause, modal-obs, nearest-cell)	worst single cheater mean +0.066; all $\leq$ the pre-registered 0.10 bar
3. Graded mixtures are monotone	Spearman $\rho = 1.000$, monotone in every cell
4. Ablation (deny forced data)	composer drops by +0.83 to +1.10
5. Shortcut-bait memorizer	memorizer $\rightarrow$ 1.000 while the composer is unchanged ($\Delta = 0.000$)
6. Retrieval+add baseline (the "is it just lookup?" attack)	pooled +0.025 $[-0.05, +0.10]$ vs. composer +0.937

Table 2: Six pre-registered validity controls. The instrument credits genuine deconfound-then-compose and rejects retrieval, memorization, and the named shortcuts. Control 6 directly answers the strongest external critique (a naive retrieval+add would tie the composer): it does not, because the prompt shows *still-confounded* forced rows, so there is no clean cell to look up—deconfounding is required.

Feasibility is decoupled from difficulty. A useful property (Table 1): the analytic ceiling stays in $[0.933, 0.938]$ *independent* of the group size C and the dial count—only the per-block row budget needed to populate the conditioning cells grows. Difficulty can therefore be raised for a model while the test remains fair at a fixed ceiling.

4 Validity of the instrument

Before using the diagnostic to make claims about models, we validate it against constructed ground truth (known composer and known shortcuts, plus a within-system ablation whose direction we know)—never against the unproven assumption "bigger model is better." Six pre-registered controls were run on 3 master seeds $\{20260605, 20260606, 20260607\}$, levels $\{2, 3, 4\}$ dials, C=5, with up to 30 puzzles per cell (fewer—16 to 17—at 2 dials, where fewer valid pairs exist). Results in Table 2.

Regime	Whole-dial (test)	All-revealed (ctrl)	Ceiling	Exact	No-data control	Strongest cheater
2 dials, C=4	+0.937	+0.937	+0.937	12/12	−0.39	+0.003
3 dials, C=3	+0.937	+0.937	+0.937	12/12	$-0.06^{\dagger}$	$+0.13^{\ddagger}$

Table 3: DeepSeek V4 Flash on the fair, gate-checked regimes. "All-revealed (ctrl)" is the positive control with i^*'s forced block *present* (same prompt format, nothing withheld); it reaches the ceiling on both regimes, so the format is solvable for this model and the whole-dial result is not a format/tally limit. "Strongest cheater" is the per-query maximum over the three cheaters, averaged over queries. †The 3-dial no-data control averages −0.06 because on one of three queries the data-free modal guess coincidentally matched the true value (+0.94); on the other two it is −0.56. This coincidence (a single-seed, 3-query artifact) is exactly why the rigorous floor comes from the multi-seed validation suite (§4), not from the LLM run's controls. ‡Driven by the nearest-cell cheater scoring +0.21/+0.18 on two queries; still $\ll$ ceiling.

5 The scale contrast

Applying the instrument to a frontier LLM (single demonstration). This is one existence-point—a single seed, a single model, and 3 held queries $\times$ 4 samples per regime—not a curve or a claim about LLMs in general. On the fair, leak-free regimes, DeepSeek V4 Flash (thinking mode, seed 20260606) reaches the analytic ceiling (Table 3): on both the 2-dial C=4 and 3-dial C=3 regimes it scores +0.937 (= ceiling) with all 12 samples (3 queries $\times$ 4) exact. An all-revealed positive control (the same format with i^*'s block *present*) also reaches the ceiling, so the instrument's format is solvable for this model; a data-free arithmetic control (same question, no data) fails, and non-compositional cheaters stay far below the ceiling—so the score reflects genuine deconfound-then-compose, not priors, memorization, or pretraining contamination.

A tiny model in the same family collapses to memorization. A from-scratch transformer ($\approx$74k parameters) with the group structure *injected* (exact circular-convolution combine) and per-dial read supervision, trained and evaluated on the same task family (n=3, C=5): with the curriculum that supervises only *present* dials' reads (forcing it to infer the withheld one), it fits the training data but the held-query withheld-effect score is −0.08; a joint-only variant scores −0.02. The independently validated architecture *can* express the answer (an all-supervised, nothing-withheld sanity reaches +0.964), so this is a learning failure, not a capacity ceiling.

Diversity is not the lever. A natural hypothesis is that the small model merely sits below a task-diversity threshold. We tested this directly (Table 4). At W=32 training worlds the model groks the training objective (training loss reaches a minimum of ≈ 0.009) yet scores ≈ 0 on *both* the withheld test *and* the easy all-revealed sanity (held queries, all blocks present). A prior report that higher diversity "stalls optimization" at W=256 was a learning-rate artifact: at lr 3×10^{-4} and 10^{-3} the training loss descends well below the uniform-prediction floor ($L_0 \approx 2.96$), to 2.03 and 1.71 respectively—optimization is alive—yet the held-query scores remain ≈ 0. Thus the model never demonstrates a generalizing read-and-compose at *any* diversity tried, even when no effect is withheld; increasing W is not what is missing.

6 Limitations

We state the limits plainly; several are load-bearing.

Worlds W	lr	Min. training loss	Withheld score	All-revealed score
32	10^{-3}	0.009	+0.016	+0.004
256	3×10^{-4}	2.03	−0.002	−0.001
256	10^{-3}	1.71	−0.010	−0.001

Table 4: Tiny-model diversity sweep (n=3, C=5, ≈74k params, 12k steps, single seed). The uniform-prediction training-loss floor is $L_0 \approx 2.96$. At W=32 the model groks training (min loss 0.009) but scores ≈ 0 on held queries including the easy all-revealed case; at W=256 optimization is alive (loss $\ll L_0$) but held-query scores stay ≈ 0. (At W=32 the loss minimum is reached mid-run; the final-step loss bounces to ≈ 0.21.)

- **Single seed, single LLM** for the headline contrast: clean points, not curves.
- **Easy, identifiable regime** (additive over a cyclic group, small C, 2–3 dials, clean noise). We found no fair regime where the LLM *breaks*; an LLM passing a known-identifiable task is expected, so the value is the contrast and the controls, not surprising the LLM.
- **The diversity-threshold defense is a-priori, not fully empirical.** Raventós et al. [4] place the threshold near 2^{14} tasks; our $W \leq 256$ is far below it. We argue—but do not empirically close—that injecting the combine operator removes the threshold's relevance: the diversity threshold governs learning the *task family*, whereas here the family (circular convolution) is *given*, leaving only the per-dial deconfounded read; the W=32-grokked-but-fails-the-easy-revealed-case control shows the model cannot even do the read it was shown, independent of diversity. A fully empirical close ($W \gg 2^{14}$, longer training) is future work.
- **Novelty rests partly on absence of evidence**—an uncovered slice in the surveyed literature, not a proof of non-existence.
- **Causal scope.** This is compositional generalization of a withheld mechanism, not Pearl–Bareinboim cross-domain transportability; we do not claim the latter.
- **Tiny-model arm:** the joint-only headline may reflect an optimization wall rather than a pure transport collapse; the read-supervised (aux-on) arm is the cleaner evidence and is what we lead with.

7 Conclusion

We contribute a small, coverage-controlled diagnostic for in-context compositional generalization of a withheld causal mechanism: it injects the known group operation to isolate transport from operation-learning, removes coverage and lookup leaks by construction, and checks fairness on the literal prompt rows at $0 before any model call. Used across scale, it makes a clean contrast legible—a frontier LLM infers the withheld mechanism at the analytic ceiling, while a tiny model given the same structure collapses to memorization and diversity does not rescue it. The phenomenon is not new; the rigor of the measurement is the point.

Reproducibility. Code and data: `https://github.com/orbitnate/prysmos-diagnostic`. All numbers above are taken from in-repo artifacts: `LLM_WHOLEDIAL_C4_2DIAL_2026-06-06.json`, `LLM_3DIAL_C3_2026-06-06.json`, `LLM_WHOLEDIAL_C4_2DIAL_REVEALED_2026-06-07.json`, `LLM_3DIAL_C3_REVEALED_`

2026-06-07.json (all-revealed positive controls), GATE0_TRAINABILITY_2026-06-06.md, RUNG1_RESULTS_2026-06-05.md, TOOL_VALIDATION_RESULTS_2026-06-05.md, and the _ramp_* sweep files; code in llm_composition_ramp.py, llm_wholedial_pilot.py, rung1_preflight.py, rung1_train.py.

References

[1] T. He, D. Doshi, A. Das, A. Gromov. *Learning to grok: Emergence of in-context learning and skill composition in modular arithmetic tasks.* NeurIPS 2024 (Oral). arXiv:2406.02550.

[2] S. Kobayashi, S. Schug, Y. Akram, F. Redhardt, J. von Oswald, R. Pascanu, G. Lajoie, J. Sacramento. *When can transformers compositionally generalize in-context?* arXiv:2407.12275.

[3] S. Lippl, K. Stachenfeld. *When does compositional structure yield compositional generalization? A kernel theory.* ICLR 2025. arXiv:2405.16391.

[4] A. Raventós, M. Paul, F. Chen, S. Ganguli. *Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression.* NeurIPS 2023. arXiv:2306.15063.

[5] A. Nguyen, G. Reddy. *Differential learning kinetics govern the transition from memorization to generalization during in-context learning.* arXiv:2412.00104.

[6] N. R. Ke, S. Chiappa, J. X. Wang, A. Goyal, J. Bornschein, M. Rey, T. Weber, M. Botvinick, M. Mozer, D. J. Rezende. *Learning to induce causal structure (CSIvA).* ICLR 2023. arXiv:2204.04875.

[7] A. Sauter, S. Salehkaleybar, A. Plaat, E. Acar. *ACTIVA: Amortized causal effect estimation via transformer-based variational autoencoder.* arXiv:2503.01290.

[8] D. Mahajan, J. Gladrow, A. Hilmkil, C. Zhang, M. Scetbon. *Amortized inference of causal models via conditional fixed-point iterations (Cond-FiP).* TMLR 2025. arXiv:2410.06128.

[9] J. Robertson, et al. *Do-PFN: In-context learning for causal effect estimation.* NeurIPS 2025. arXiv:2506.06039.

[10] F. Montagna, M. Cairney-Leeming, D. Sridhar, F. Locatello. *Demystifying amortized causal discovery with transformers.* TMLR 2025. arXiv:2405.16924.

[11] N. Schneider, L. Lorch, N. Kilbertus, B. Schölkopf, A. Krause. *Generative intervention models for causal perturbation modeling (GIM).* arXiv:2411.14003.

[12] A. Abedsoltan, H. Zhang, K. Wen, H. Lin, J. Zhang, M. Belkin. *Task generalization with autoregressive compositional structure: Can learning from D tasks generalize to D^T tasks?* arXiv:2502.08991.